

From Clefts to Definiteness in Bangla and Beyond[‡]

ABSTRACT

I provide empirical support for the claim that the definiteness of a DP can be decomposed into an existential quantifier and a cleft operator attached to the restrictor of this quantifier. As empirical evidence, I provide novel data from Bangla, where exactly the kind of DPs that otherwise don't allow a definite reading suddenly allows a definite reading when a cleft morpheme is put on top of them. I propose that the standard uniqueness/maximality presupposition of definites is derivable as the trivalence resulting from this cleft operator inside the restrictor of an existential, projecting to the matrix level according to the Strong Kleene schema. Additional data is presented from English bearing on the separability of the cleft operator and the indefinite determiner in definites in general. Moreover, in addition, my account predicts weaker, conditional presuppositions for definites, which I argue to be detectable, when appropriate contexts are provided. I argue that the standard uniqueness/maximality presupposition fails to account for these cases. The standard uniqueness/maximality presupposition, I suggest, can be derived from the presupposition my account predicts via proviso-like strengthening.

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1 Introduction

It is an understatement to say that very little is well-understood when it comes to the distinction between definiteness and indefiniteness. There are seminal works (Frege 1952, Link 1984, Russell 1905, Sharvy 1980, Strawson 1950) and more recent works on this (e.g., Bumford 2017a,b, 2018, Coppock & Beaver 2012a,b, 2014, 2015, Heim 1982, among many many others) that approach this issue from a variety of theoretical perspectives. With all this in the background, it is by now intuitive to people familiar with the subject that, on a certain relevant level of abstraction, indefinites introduce novel discourse references, which is known

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as *the Novelty Condition*, and definites refer back to already introduced discourse referents, which is known as *the Familiarity Condition*. For instance, in (1), the speaker must not be referring to Mary by *a woman*; it has to be someone that the discourse has not yet mentioned. But, in (2), it is the exact opposite: John must be referring to someone familiar to both him and his interlocutors by *the woman*.

- (1) I was talking to Mary, when a woman suddenly sneezed.
 - a. **Context:** The speaker is talking to Mary. Sue is also in the room during their conversation. The speaker does not know Sue. In the middle of their conversation, Sue suddenly sneezes, startling the speaker.
 - b. **#Context:** The speaker is talking to Mary. There is no one else around them. In the middle of their conversation, Mary suddenly sneezes.

- (2) The woman probably has a cold.
 - a. **Context:** John and Bill are sitting a bit far away from a woman who has been coughing for a long time. There is no one else around. John says this to Bill.
 - b. **#Context:** John and Bill have an appointment. On his way to the appointment, John sees a woman who is coughing a lot. Then she goes in a different direction and he does not see her again. Bill has never seen this woman, does not know she exists. John comes to the meeting and utters the sentence to Bill.

Moreover, the definite determiner is usually described to come with a uniqueness/maximality presupposition. For instance, if there is more than one book on a table, it is infelicitous to say *The book is red*, and if there are more than two books on a table, it is infelicitous to say *The books are red* with an intended meaning that only some of the books on the table are red; *the books*, there, can only pick out the maximal set of books. This is summarized in (3)-(4).

- (3) The book is red.
 - a. **Context:** There is exactly one book on a table.
 - b. **#Context:** There are multiple books on a table.

- (4) The books are red.
 - a. **Context:** There are multiple books on a table, all of which are red.
 - b. **#Context:** There are multiple books on a table, only some of which are red. The remaining are blue.

These facts are conventionally taken to point to a fundamental difference between the lexical meaning of the indefinite determiner and that of the definite determiner, which is commonly

done via a presupposition that definite determiners have to satisfy when they combine with their NP arguments: which is that its NP argument must denote a singleton set of objects matching its description. This can be captured with a naïve entry for the definite determiner like (5). I am following the Heim & Kratzer (1998) convention of indicating the presupposition of a lexical item by putting it after a colon.

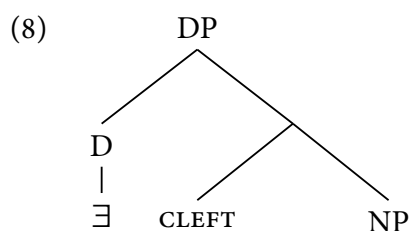
$$(5) \llbracket \text{the} \rrbracket = \lambda P_{\langle e, t \rangle} : \exists! x P(x) = 1 . \iota y P(y) = 1$$

On the other hand, the indefinite determiner can be given the following lexical entry, which can take a set of objects to yield a generalized existential quantifier. Crucially, it does not come with any presupposition.

$$(6) \llbracket a \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \exists x [P(x) = Q(x) = 1]$$

In this paper, I want to provide evidence for the claim that the difference between definites and indefinites is more minimal than these initial considerations would make us believe. This proposal has already appeared as a theory of definites in Doron (2025). The purpose of this paper is to bolster this theory with evidence. At the heart of this proposal here is the claim that definites always have an indefinite part, and something extra gets added to the existential meaning of this indefinite, and these two things together create the inference of uniqueness/maximality.¹ This extra thing, I want to argue, is exactly that operator that gives rise to the presuppositional nature of clefts. By cleft, I mean sentences of the form in (7). I specifically propose (8) as the internal structure of a definite, where CLEFT stands for the cleft morpheme (more on this below).

(7) It's John that we invited.



This idea is not new. At least the intuition that some amount of separation is needed between the various components of the meaning of the definite determiner has already appeared, for instance, in the work by Bumford cited above. However, the specific idea I want to propose is more radical than what has been proposed before. I'm proposing, not just that there are

1. In Doron (2025), the exhaustifying operator is present in both indefinites and definites and the difference between them arise through a difference in focus association. I want to remain agnostic about whether this is necessary. All I want to focus on here is definiteness arising from this exhaustifying operator. The particular view I have in this paper can be reconciled Doron's broader approach.

two separate things in the meaning of the definite determiner, but that this other component is specifically the element independently responsible for the semantics of cleft, that they are separable, detachable, in a meaningful way, and the effect of this detachment is empirically detectable. As mentioned above, [Doron \(2025\)](#) has already developed this idea as a theory of definiteness, which my paper is inspired by. His contribution there is mostly theoretical. One crucial goal of this paper is to furnish empirical motivation for such a theory, that is, to argue that definites *have* to be treated in this way.

To put it in more concrete terms, I am advocating a theory under which the uniqueness/maximality presupposition of definites can be derived from an indefinite whose restrictor cleft morpheme. This cleft, being a presupposition trigger, turns the restrictor of the indefinite into a partial function, and the global consequence of this partiality is what is perceived as uniqueness/maximality. Specifically, I will assume a trivalent framework of presupposition projection ([Beaver & Krahmer 2001](#), [Fox 2008, 2013](#), [George 2008a,b](#), [Sudo et al. 2012](#)). The cleft operator will be a presupposition trigger, that is, it will output trivalent semantic objects. This trivalence will make the restrictor of the indefinite partial. Assuming a Strong Kleene schema of presupposition projection, this trivalence will project to the matrix level from the restrictor of the indefinite, and resulting global trivalence will be equivalent to the uniqueness/maximality presupposition standardly attributed to the definite determiner.

On the empirical front, I want to present novel data from Bangla in this paper which I believe provide evidence for this way of thinking about definiteness. In short, Bangla DPs that have the NUMERAL NP order are indefinites, while DPs that have the NP NUMERAL order (NP-first DPs) are definites. But when a cleft morpheme is attached to numeral-first DPs, these numeral-first DPs, now bearing the cleft morpheme, become ambiguous between a definite reading (despite their NUMERAL NP order) and another reading that can be described as a cleft that has the indefinite meaning of the numeral-first DP as its pivot. This pattern from Bangla provides evidence for Doron's analysis, given the source of cleft semantics has been independently analyzed as an effect of trivalent exhaustification ([Fox 2020](#), building on [Büring & Križ 2013](#)). This gives us reason to rethink the syntax-semantics of definite DPs cross-linguistically.

In the last part of the paper, I present evidence to the same effect from English and, specifically, to the separability of the cleft morpheme from the indefinite determiner. This evidence comes from cases of apparent domain restriction, where, under the conventional theory of definiteness, it must be explained why a sentence comes out as felicitous despite the fact that there is a definite determiner inside it whose uniqueness/maximality presupposition is not met. To preview the facts briefly, it is possible to say *John picked the red ball* even when there are multiple red balls and balls of other color salient in the context, but John is the only individual who picks up a red ball; that is, the use of the definite determiner is licensed only with respect to a smaller set of balls, that is, the set of balls picked by John and some other contextually salient individuals. This can be explained away by restricting the domain of the definite determiner to this smaller set (and this can happen through whatever happens to be one's favorite way to implement domain restriction). However, following [Bumford \(2018\)](#), who made similar arguments in a slightly different context, I show

that such a domain restriction view is inadequate. Instead, allowing the cleft morpheme to move out of the indefinite naturally resolves the issue. Crucially, such an account relies on the separability of the cleft morpheme from the indefinite determiner, which is the cornerstone of my proposal.

2 The Bangla facts

2.1 Background: (in)definiteness in Bangla

Since this paper is partly concerned with the behavior of Bangla numeral-modified DPs, I will focus on the (in)definiteness properties of only such DPs in the language. One prominent way the distinction between indefinite and definite numeral-modified DPs is marked in Bangla is word order. This is shown in (9). In indefinite DPs, the numeral precedes the NP restrictor, as in (9a), while, in definite DPs, the numeral follows the NP restrictor, as in (9b).²

- (9) a. ami **du-to** **boi** poɽ-etʃʰ-i.
 1.SG.NOM **two-CLF** **book** read-PERF-PRS.1
 ‘I’ve read two books.’
- b. ami **boi** **du-to** poɽ-etʃʰ-i.
 1.SG.NOM **book** **two-CLF** read-PERF-PRS.1
 ‘I’ve read the two books.’

Before I get to the difference between definiteness and indefiniteness that becomes apparent in fuller, more plausible discourses,³ I will first introduce some assumptions about definite and indefinite DPs with numerals in English. The Novelty condition and Familiarity Condition discussed in the introduction also manifests when definites and indefinites have numerals in them. This is illustrated in (10) for the Novelty Condition, and (11) for the familiarity condition. The two indefinites in (10) both introduce novel discourse referents, which is why they are acceptable in Context 1. But the second indefinite cannot refer back to the discourse referents introduced by the first one, which is why the discourse is unacceptable in Context 2. In (11), the definite *the two students* must refer back to an already introduced discourse referent, which is why it is acceptable in Context 3. But it is unacceptable in Context 4, because, there, it has to refer to individuals who have not been introduced in the discourse, that is, Bill, and Sue.

2. All Bangla glosses are given in the IPA and glossing abbreviations are fully spelled out at the end of the paper.

3. Discussion of simpler examples showing such definite-indefinite alternation can be found in [Bhattacharya \(1998, 1999\)](#), [Dayal \(2012, 2014\)](#).

- (10) **Two students** wrote a paper on Benjamin; **two** on Adorno.
- a. **Context 1:** There are four students in a philosophy class. Two of them write their term papers on Benjamin, and the other two, on Adorno.
 - b. #**Context 2:** There are two students in a philosophy class. Both of their terms papers had to do with Benjamin and Adorno.
- (11) John and Mary wrote their papers on Benjamin. **The two students** find Adorno more difficult.
- a. **Context 3:** John and Mary are two students in a philosophy class. They both find Adorno more difficult than Benjamin. So, they write their term paper on Benjamin.
 - b. #**Context 4:** John, Mary, Bill, and Sue are students in a philosophy class. John and Mary find Benjamin more difficult than Adorno. Bill and Sue find Adorno more difficult than Benjamin.

The same pattern can be seen in Bangla. (12) features the numeral-first order and (13), the NP-first order. What (12) shows is that, once two students have been introduced through an indefinite, another same/similar indefinite obligatorily introduces two other students and fails to refer back to the first two students. This is why the discourse is acceptable in Context 1, but unacceptable in Context 2. Therefore, the numeral-first order lacks a definite meaning. (13) shows that, when two students have been introduced, a DP with an NP-first order in later discourse obligatorily has a definite meaning, that is, must refer back to the two students previously introduced, and fails to introduce two other students. This is why the discourse is acceptable in Context 3, but unacceptable in Context 4. Therefore, the NP-first order has a definite reading.⁴

- (12) **du-ḡon student** benjamin-er upor pepar lik^h-l-o; ar **du-ḡon**
two-CLF student Benjamin-GEN on paper write-PFV-PST.3 and **two-CLF**
 æḡomo-r upor.
 Adorno-GEN on

“Two students wrote a paper on Benjamin; and two on Adorno.”

“Two students wrote a paper on Benjamin; and those two students wrote a paper on Adorno.”

- a. **Context 1:** There are four students in a philosophy class. Two of them write their term papers on Benjamin, and the other two, on Adorno.
- b. #**Context 2:** There are two students in a philosophy class. Both of their terms papers had to do with Benjamin and Adorno.

4. As far as deriving the variation in the NUMERAL NP order is concerned, Bhattacharya (1998, 1999) posits DP-internal NP movement past the numeral, which is observed only in definite DPs. The properties of and the constraints on this movement aren't immediately relevant to the purposes of this paper.

- (13) *dmol ar tʃaru benjamin-er upor pepar lik^h-l-o.*
 Amol.NOM and Charu.NOM Benjamin-GEN on paper write-PFV-PST.3
student du-to-r ædono beʃi koʃin lag-e.
 student two-CLF-GEN Adorno much difficult feel-HAB.PRS.3
 “Amol and Charu wrote their papers on Benjamin. Those two students find Adorno more difficult.”
 # “Amol and Charu wrote their papers on Benjamin. The two other students find Adorno more difficult.”
 a. **Context 3:** Amol and Charu are two students in a philosophy class. They both find Adorno more difficult than Benjamin. So, they write their term paper on Benjamin.
 b. # **Context 4:** Amol, Charu, Monda, and Uma are students in a philosophy class. Amol and Charu find Benjamin more difficult than Adorno. Monda and Uma find Adorno more difficult than Benjamin.

(3) further confirms that numeral-first DPs have no uniqueness/maximality presupposition, while NP-first DPs do.

- (14) **Context:** Charu, Monda, and Aparna (all of whom are women) are ill.
- a. *du-ɕon mofila ɔjust^ho. bɔl-a b^halo, tin-ɕon mofila ɔjust^ho.*
 two-CLF woman ill say-NMLZ good three-CLF woman ill
 “Two women are ill. It’s better to say that three women are ill.”
- b. *tin-ɕon mofila ɔjust^ho.*
 three-CLF woman ill
 “Three women are ill.”
- c. # *mofila du-ɕon ɔjust^ho.*
 woman two-CLF ill
 # “The two women are ill.”
- d. *mofila tin-ɕon ɔjust^ho.*
 woman three-CLF ill
 “The three women are ill.”

I will, therefore, take this set of effects as evidence in favor of the claim that DPs with the NUMERAL NP order are indefinite and those with the NP NUMERAL order are definite.

2.2 Introducing [-i]

Enter [-i]. [-i] is a focus-sensitive contrastive focus-marking morpheme in Bangla. In this section, I will explain how [-i] interacts with a sentence. (15) exemplifies a fairly common

usage of this morpheme. Here, for instance, (15a) does not establish the speaker's steadfast determination to wear the dress in question, as opposed to other pieces of clothing. So, one could say this sentence to express a preliminary choice of clothing which they might be willing to change, given other options. But, in (15b), the sentence conveys that the speaker is determined to wear the dress that is spoken about in the sentence, and nothing else. That is, a contrast is established between the dress in question and other pieces of clothing; nothing will change their mind now.

- (15) a. ami ei ḡama-ṭa por-b-o.
 1.SG.NOM this dress-CLF wear-FUT-1
 “I will wear this dress.”
- b. ami ei ḡama-ṭa-i por-b-o.
 1.SG.NOM this **dress-CLF-FOC** wear-FUT-1
 “It’s this dress that I’ll wear. (Not anything else.)”

Coming back to numeral DPs now, just like “dress-CLF” in (15b), this morpheme can attached to anything, including numeral-modified DPs which are, for instance, numeral-first in order. (16) is such an example. What is crucial, though, is the divergence of the behavior of this numeral-first DP with an *[-i]*, from that of the exact same DP without the morpheme, in (12). While, in (12), a definite reading of the DP is impossible, in (16), a definite reading — referring back to the two books previously introduced — becomes surprisingly available, which is the reading made salient by (16b). Throughout the rest of the paper, I will call the first reading *the cleft reading of [-i] DPs*, and the second reading, *the definite reading of [-i] DPs*. The cleft reading of (16) is completely parallel to the meaning of (15): here, the number of books that were found is being contrasted with higher numbers.

- (16) o ḡu-ṭo boi-i k^hũḡ-e pe-et^h-e.
 3.SG.NOM **two-CLF** **book-FOC** find-GER get-PERF-PRS.3
 “He has found TWO books.”
 “He has found the two books.”
- a. **Context 1 (the cleft reading):** Amol has found some books to add to the optional reading list for his class. He was looking for two more appropriate books to add. He was having a hard time finding such books. So, he considers adding just one more book, which he manages to find, and tells the speaker about this change in plans, without saying which book he found. So, the speaker starts to believe that Amol has found only one book. But, in the course of the next few days, he comes across a very nice book which he would definitely add and happily reports that to the speaker. The speaker now does not know which books Amol has finally got, but only that Amol has found two books that he can now add to his list of optional readings. So, he corrects his beliefs.

- b. **Context 2 (the definite reading):** Amol and Charu are siblings and go the bookstore together to buy books. Amol gets two books. Charu is present during the purchase. She sees those books, and when they come home, their mother sees the books as well. After that, Amol is seen reading them for a few days. Then Amol loses the books. Amol, Charu, and their mother look for it together. After a long search, Amol manages to find them and Charu goes to their mother to report that, so she can stop looking.

For the sake of completeness, I should point out that, in the cleft reading of (16), made salient by Context 1, instead of the number of books, the fact that they are books as opposed to something else could also be contrasted by putting focus on *book*. In that case, the meaning that would arise would be *He has found two BOOKS, not two LAPTOPS/PENS/. . .*. This variation is relevant and possible every time the cleft reading is possible. Since my purposes here concern just the possibility of the cleft reading — and not the variations thereof because of different focus structures — I will set aside the possibility of this variation throughout this paper and talk as if only one kind of cleft reading is there, one where the numeral is focused. But that is only for the sake of exposition.

(17) is an example where the definite reading is more obvious. (17a) shows that the numeral-first order is not acceptable, where the cardinality of the noun's denotation is exactly the same as the numeral's denotation. That is, it behaves exactly like the indefinite *two of my legs*, which is predicted by Maximize Presupposition. (17b) shows that the NP-first order is acceptable, and this can be made sense of, because the NP-first order signals definiteness. Crucially, (17c) shows that numeral-first [-i] DPs are acceptable in the same context, which can only be explained if they have a reading that behaves like the definite in (17b), and not like the indefinite in (17a). Therefore, this provides explicit evidence in favor of the definite reading of numeral-first [-i] DPs.

(17) **Context:** The speaker has broken both of their legs.

- a. # ama-r **du-to** pa b^hej-e gætʃ^h-e.
 1.SG-GEN **two-CLF** leg break-GER go-PERF-PRS.3
 # “Two of my legs have broken.”
- b. ama-r **pa du-to** b^hej-e gætʃ^h-e.
 1.SG-GEN **leg two-CLF** break-GER go-PERF-PRS.3
 “My two legs have broken.”
- c. ama-r **du-to** **pa-i** b^hej-e gætʃ^h-e.
 1.SG-GEN **two-CLF** **leg-FOC** break-GER go-PERF-PRS.3
 “My two legs have broken.”

3 The compositional semantics of the cleft reading

3.1 Conditional presupposition

The goal, at the end of the day, is to make sense of (16). That will be possible when it is clear how the semantics of *[-i]* works independently. This is what I will do in this section. I will begin by introducing some assumptions about the semantics of *[-i]*. I will do this by focusing on (15), repeated below. I will begin with this because it does not have the ambiguity seen in (16); so it will help us understand the usual profile of *[-i]*.

- (15) ami ei **ḡama-ta-i** por-b-o.
 1.SG.NOM this **dress-CLF-FOC** wear-FUT-1
 “It’s this dress that I’ll wear. (Not something else.)”

The goal of this section is to argue that sentences of this form and meaning have a presuppositional component in their interpretation which is very similar to the presuppositional nature of clefts. In order to do that, I will first point out some properties of clefts that have been independently observed in the literature in the context of English. I will then apply those considerations to sentences like (15) in Bangla, which will reveal that clefts and sentences with *[-i]* are semantically equivalent.

To keep things simple and not get into syntactic issues, I will assume, following [Velleman et al. \(2012\)](#), that the cleft construction involves introducing a cleft operator — let us call it *CLEFT* — into the structure. [Büiring & Križ \(2013\)](#) have proposed a presupposition for clefts, which [Fox \(2020\)](#) has stated as a conditional/disjunctive one. For instance, (18a) has the presupposition in (18b). [Büiring & Križ](#) establish, through various examples, that neither a simple exhaustivity presupposition (*She invited only Fred*) nor a singularity presupposition (*She invited exactly one person*) accurately captures this inference we seem to get from clefts.

- (18) a. It was Fred that she invited. ([Büiring & Križ 2013](#), (9): 4)
 b. **Presupposes:**
 If she invited Fred, then she invited only Fred.
 Equivalently:
 Either she didn’t invite Fred, or she invited only Fred.

I will begin by briefly reproducing [Büiring & Križ’s \(2013\)](#) reasons for adopting this presupposition. Their point of departure is the observation that the exhaustivity inference of a cleft cannot be part of its assertoric component. This can be seen in (19). When a cleft is negated for the sole reason to convey that the exhaustivity claim is false, the result is infelicitous, as shown in (19a). This is to be contrasted with the behavior of *only*, given in (19b), which allows such negation because the exhaustivity information is in the assertoric component of the meaning of *only*.

- (19) a. #She invited Fred, but it wasn't Fred she invited.
b. She invited Fred, but she didn't invite only Fred.

(Büiring & Križ 2013, (4): 2)

Since the exhaustivity inference cannot be part of the assertion of a cleft, the next possibility is that it is a presupposition. Obviously, (18a) cannot presuppose *She invited only John*. Otherwise, it would not be possible to make sense of why it is felicitous to ask *Is it John that we invited?*. The other possibility is the singularity presupposition, that is, *We invited exactly one person*. But that cannot be right either, because that should make the discourse in (20) infelicitous. This is because, if the cleft in the first sentence presupposes *She invited exactly one person*, then it should not have been felicitous to add that she invited more than one person, that is, Bob and Gord.

- (20) It wasn't Fred she invited. She invited Bob and Gord.

(Büiring & Križ 2013, (6): 3)

Büiring & Križ's (2013) solution to this is to adopt the conditional presupposition in (18b).⁵ Once this is adopted, the exhaustivity inference of an affirmative cleft like (21) follows, as in (21c) from the conjunction of the assertion ((21a)) and the presupposition ((21b)). In the case of a negated cleft like (22), the conjunction of the assertion ((22a)) and the presupposition ((22b)) does not give rise to any exhaustivity or singularity inference. This is because the assertion makes the antecedent of the conditional false. This explains why (20) is felicitous: if Fred was not invited, then, obviously, the consequent cannot be true, and also, it could be that more than one person was invited and Fred just was not one of them. Based on these reasons, I will adopt the conditional presupposition for clefts.

- (21) It was Fred she invited.
a. **Assertion:** She invited Fred.
b. **Presupposition:** If she invited Fred, then she invited only Fred.
c. $\rightsquigarrow$ She invited only Fred.

5. They do not state their adopted presupposition in a conditional form, strictly speaking. Their proposed presupposition for (18a) is the following: *Fred is not a proper part of the sum of all people invited by her*. The reason I do not adopt this formulation involving parthood is that the notion of parthood that seems to be relevant for (18a) is just an effect of an entailment relationship between alternatives. For instance, in the case of the sentence *It's two packets of flour that will be sufficient*, two packets of flour are indeed a proper subpart of the amounts of flour that will be sufficient, because, if two packets of flour are sufficient, then so are three, four, five, or six of them, and so on. This is a clear indication that parthood is not at stake, entailment relations between propositions are. The spirit of this argument has been elaborated in the context of maximal informativity in definites in von Stechow et al. (2014).

(22) It wasn't Fred she invited.

- a. **Assertion:** She did not invite Fred.
- b. **Presupposition:** If she invited Fred, then she invited only Fred.
- c. $\neg \rightarrow$ She invited only Fred.
 $\neg \rightarrow$ She invited exactly one person.

Coming back to (15), I argue that the *[-i]* morpheme in Bangla is equivalent to CLEFT in English. I argue this on the basis of identical behavior of Bangla to the English (19) and (20). In (23), when a sentence with *[-i]* is uttered for the sole reason of conveying the exhaustivity inference, the result is infelicitous. In (24), if singularity was a presupposition of the sentence with *[-i]*, then that would have been the presupposition of A's questions, which would have made B's response infelicitous.

(23) #*dmol* *ʒan-t-o* *ʒe* *ami* *ei* ***ʒama-ta*** *por-b-o.* *aʒ*
 Amol.NOM know-HAB-PST.3 that 1.SG.NOM this **dress-CLF** wear-FUT-1 today
ʒan-l-o *ʒe* *ami* *ei* ***ʒama-ta-i*** *por-b-o.*
 know-PFV-PST.3 that 1.SG.NOM this **dress-CLF-FOC** wear-FUT-1

Intended:

#“Amol knew that I would wear this dress. Today, he found out that it's this dress that I'll wear.”

(24) A: *dmol* *ki* ***ʃaru-r*** ***ʃa^he-i*** *af-b-e?*
 Amol-NOM Q_{Y/N} **Charu-GEN** **with-FOC** come-FUT-3
 “Is it Charu that Amol will come with?”
 B: *n-a,* *o* *uma* *ar* *mōnda-r* ***ʃa^he*** *af-b-e.*
 NEG-IMPV 3.SG.NOM Uma and Monda-GEN with come-FUT-3
 “No, he will come with Uma and Monda.”

I will, therefore, assume that *[-i]* is responsible for a meaning that is identical to English clefts. That is, (15), repeated in (25), asserts (25a) and presupposes (25b). These two, in combination, yield the exhaustivity inference.

(25) *ami* *ei* ***ʒama-ta-i*** *por-b-o.*
 1.SG.NOM this **dress-CLF-FOC** wear-FUT-1
 “It's this dress that I'll wear. (Not something else.)”
 a. **Assertion:** I will wear this dress.
 b. **Presupposition:** If I wear this dress, then I wear only this dress.

Formally, I will assume a trivalent framework of presupposition projection (Beaver & Krahmer 2001, Fox 2008, 2013, George 2008a,b, Sudo et al. 2012). In this framework, the meaning

of a sentence is not formally bifurcated into an assertoric component and a presuppositional component. Instead, the truth and falsity conditions of a sentence are defined in such a way that it may not be either true or false in certain cases. If the truth and falsity conditions are complementary to each other, then the sentence always has a classical truth value (0 or 1). If they are not complementary to each other, then that means that there are worlds in which the sentence is neither true nor false. In those worlds, it receives a third truth value, which expresses what is conventionally understood as undefinedness because of presupposition failure. In this framework, the definedness/felicity condition of a sentence, therefore, is the disjunction of its truth condition and its falsity condition. This disjunction is equivalent to what is conventionally described as the presupposition of the sentence. And this generalizes to expressions of all types, not just sentences. That is, the truth and falsity conditions of each node in a structure can be computed from bottom up; so their disjunctions can be computed, and the presupposition of every node can, therefore, be computed in this way.

Following Fox (2020), I will encode the meaning of (15) as in (26) in this trivalent framework, with the truth and falsity conditions as shown. When the disjunction of these two conditions are taken, it turns out to be *Either the speaker does not wear the dress involved or the speaker wears only the dress involved*. Because of *modus tollens*, this is equivalent to the desired conditional presupposition *If the speaker wears the dress involved, then the speaker wears only the dress involved*.

$$(26) \quad \llbracket \text{ami e}\bar{\text{i}} \text{ } \check{\text{d}}\text{ama-}[\text{a-}\bar{\text{i}} \text{ por-b-o}] \rrbracket \\ = \begin{cases} 1 & \text{if the speaker wears the dress involved and nothing else} \\ 0 & \text{if the speaker doesn't wear the dress involved} \\ \# & \text{otherwise} \end{cases}$$

These truth conditions will fall out from the lexical entry for $[-i]$ given in (27). For completeness, I would give the same lexical entry for CLEFT in English as well. For now, let us say that $[-i]$ and CLEFT can attach to t -type nodes. This is indicated by suffixing “ st ” to $[-i]$. Regarding its pronunciation adjacent to the pivot/focus associate (more on this in the next section), I propose that it originates on the pivot and moves to the t -type nodes in narrow syntax; then, at LF, the lower copy is deleted, and, at PF, the higher copy is deleted. For the purposes of this paper, I would adopt a simple version of the exhaustifying operator EXH, which simply negates stronger alternatives. Notice that I am using the EXH operator in the metalanguage, but its metalanguage function is described by putting object language in double brackets. So, “ $\llbracket \text{EXH}^{st} \phi \rrbracket$ ” should be understood as the denotation of the result of combining EXH and ϕ , which would happen via Function Application. In a sense, the meaning is written by referring to a possibly hypothetical EXH operator in the object language, and this is done for ease of exposition and conceptualization. Whether EXH exists in the object language separately is orthogonal to my purposes here and I would like to remain agnostic about it. To reduce clutter, I omit C (the set of alternatives) in (27) (and wherever else it does not play a crucial

role).^{6, 7}

$$\begin{aligned}
 (27) \quad & \llbracket \neg^{st} \phi \rrbracket \\
 &= \begin{cases} 1 & \text{if } \llbracket \text{EXH}^{st} \phi \rrbracket = 1 \\ 0 & \text{if } \llbracket \neg \phi \rrbracket = 1 \\ \# & \text{if } \neg(\llbracket \phi \rrbracket = 1 \rightarrow \llbracket \text{EXH}^{st} \phi \rrbracket = 1) \end{cases} \\
 &\rightsquigarrow \begin{cases} \text{prs: } \llbracket \phi \rrbracket = 1 \rightarrow \llbracket \text{EXH}^{st} \phi \rrbracket = 1 \\ \text{asr: } \llbracket \phi \rrbracket \end{cases}
 \end{aligned}$$

$$(28) \quad \llbracket \text{EXH}^{st} \rrbracket^w(C)(p) = 1 \text{ iff } p(w) = 1 \wedge \forall q \in C[p \not\subseteq q \rightarrow q(w) = 0]$$

3.2 The pivot

In this section, I will make explicit what I mean by the *pivot* of a cleft. As [Velleman et al. \(2012\)](#) correctly point out, [Büring & Križ \(2013\)](#) assume too strong a relationship between cleft syntax and cleft semantics. That is, of course, it makes sense, from the point of view of syntax, to call the entire bolded phrase in (29) — the head of the relative clause — *the pivot* of the cleft. After all, this is what undergoes syntactic movement, which is, in this case, relativization. However, this phrase is not coextensive with the focus associate of CLEFT, which is a separate semantic notion. If it were, then only the meaning in (29a) would arise, the ones in (29b) and (29c) wouldn't: in (29b), the adjective *eldest* is the focus associate, while, in (29c), *John* is the focus associate; but what moves in these cases is something that properly contains them.

- (29) It was **John's eldest daughter** who liked the movie.
- a. It was [John's eldest daughter]_F who liked the movie.
 $\rightsquigarrow$ No other people liked the movie.
 - b. It was John's [eldest]_F daughter who liked the movie.
 $\rightsquigarrow$ None of John's other daughters liked the movie.
 - c. It was [John's]_F eldest daughter who liked the movie.
 $\rightsquigarrow$ Nobody else's eldest daughters liked the movie.

([Velleman et al. 2012](#), (1): 442)

6. One could adopt more sophisticated meanings for EXH, for instance, one that is sensitive to Innocent Exclusion ([Fox 2007](#)), but it does not matter for my purposes; so I have kept it as simple as possible.

7. Sometimes, I will restate trivalent meanings as the combination of the resulting presupposition ("prs") and the assertion ("asr"), to make the semantics easy to understand.

This distinction is not quite appreciated in Buring & Križ (2013). Instead, they take the whole syntactically moved phrase to be the focus associate of (their equivalent for) CLEFT. However, Velleman et al. (2012) also point out that Buring & Križ's (2013) proposed conditional presupposition fares as well as their account, except for the issue just discussed. I will, therefore, assume the conditional presupposition (anticipating the way it allows us to see the predictions I want to discuss), but implement it more along the lines of Velleman et al. (2012). That is, semantically, the associate of CLEFT is only whatever needs to be focused, not the whole syntactic pivot. I will call this *the semantic pivot of the cleft* or *the focus associate of the cleft*, interchangeably. Therefore, for the semantic pivots that are seen in (29), the following LFs in (30) can be given. For the purposes of this paper, I will ignore the issue of the movement and pronunciation aspect of this LF, that is, how the relativization movement and the pronunciation of the syntactic pivot to the left of the relative clause happens, even though the focus associate is inside the syntactic pivot.

- (30) It was **John's eldest daughter** who liked the movie.
- a. It was [John's eldest daughter]_F who liked the movie.
 $\rightsquigarrow$ No other people liked the movie.
LF: [CLEFT [[JOHN'S ELDEST DAUGHTER]_F liked the movie]]
 - b. It was John's [eldest]_F daughter who liked the movie.
 $\rightsquigarrow$ None of John's other daughters liked the movie.
LF: [CLEFT [John's [ELDEST]_F liked the movie]]
 - c. It was [John's]_F eldest daughter who liked the movie.
 $\rightsquigarrow$ Nobody else's eldest daughters liked the movie.
LF: [CLEFT [[JOHN'S]_F eldest daughter liked the movie]]

4 Where is [-i] in the definite reading?

The preceding discussion on CLEFT and [-i] serves to provide a sense of what will be the analysis of the cleft reading of numeral-first [-i] DPs in Bangla: basically it is a cleft where the focus associate of the CLEFT operator is an indefinite DP with a numeral in it. What is now left to be accounted for is how the definite reading of numeral-first [-i] DPs arises. In order to do that, it needs to be understood how [-i] interacts with the rest of the material inside the DP. This is what I will do now.

On the one hand, as I have argued above, numeral-first [-i] DPs are ambiguous between a cleft reading and a definite reading. On the other hand, NP-first DPs without [-i] — which requires DP-internal NP movement past the numeral — only have a definite reading, as shown in (12) and (13). This can be summarized as the following table:

	WITHOUT [-i]	WITH [-i]
NUMERAL-FIRST	indefinite	cleft with an indefinite pivot definite
NP-FIRST	definite	??

The “??” stands for the fact that I have not yet shown what happens when the [-i] morpheme is put on an NP-first DP: does its usual definite reading stay intact? Does some other reading arise? Does the definite reading of numeral-first [-i] DPs arise? (31) seems to answer in the negative to this question. If NP-first [-i] DPs had the same definiteness meaning as the definite reading of numeral-first [-i] DPs, then there should not have been any difference between (31a) and (31b). Instead, while the former is acceptable, the latter is not; and it feels that, in (31b), the “wrong” thing is being focused because focusing [pa du-to] “leg two-CLF” requires contextual saliency of something other than the sum of the two legs. The fact that this is indeed what goes wrong in (31b) is confirmed by the felicity of (31c), where this factor is controlled for by providing the context with such an alternative thing could be broken, that is, the hands. I have not given any translation for (31b) because there is no good translation for it. The point of (31b) is simply to show that it is different from (31a).

- (31) a. ami ḡan-t-am ḡe ḡmol-er ḡntoto æk-ṭa pa
 1.SG.NOM know-HAB-PST.1 that Amol.GEN at.least one-CLF leg
 b^{hi}ej-et^h-e. aḡ ḡan-l-am ḡe o-r du-to
 break-PERF-PRS.3 today know-PFV-PST.1 that 3.SG.GEN two-CLF
 pa-i b^{hi}ej-et^h-e.
 leg-FOC break-PERF-PRS.3

“I knew that Amol broke at least one of his legs. Today, I found out that he has broken both of his legs.”

- b. #ami ḡan-t-am ḡe ḡmol-er ḡntoto æk-ṭa pa
 1.SG.NOM know-HAB-PST.1 that Amol.GEN at.least one-CLF leg
 b^{hi}ej-et^h-e. aḡ ḡan-l-am ḡe o-r pa
 break-PERF-PRS.3 today know-PFV-PST.1 that 3.SG.GEN leg
 du-to-i b^{hi}ej-et^h-e.
 two-CLF-FOC break-PERF-PRS.3

- c. ami b^{hi}eb-et^h-il-am ḡe ḡmol-er haṭ du-to
 1.SG.NOM think-PERF-PFV-PST.1 that Amol.GEN hand two-CLF
 b^{hi}ej-et^h-e. kiṇtu aḡ ḡan-l-am ḡe o-r pa
 break-FUT-3 but today know-PFV-PST.1 that 3.SG.GEN leg
 du-to-i b^{hi}ej-et^h-e.
 two-CLF-FOC read-PERF-PST.3

“I thought that Amol had broken his hands. But today I found out that it’s his legs that he has broken.”

This allows us to update our table as below:

	WITHOUT $[-i]$	WITH $[-i]$
NUMERAL-FIRST	indefinite	cleft with an indefinite pivot definite
NP-FIRST	definite	cleft with a definite pivot

Therefore, it can be concluded that the definite reading of numeral-first $[-i]$ DPs cannot coexist with the definite reading of NP-first DPs (with or without $[-i]$). This would only make sense if attaching $[-i]$ where it needs to be attached in order for the definite reading to arise prevents the definiteness-sensitive DP-internal NP movement, because the above example shows that these two things cannot coexist. That means that, the LF that the definite reading of numeral-first $[-i]$ DPs arises from must have $[-i]$ inside the DP.

5 A solution: trivalent exhaustification of restrictors of DPs

To summarize: I have shown that there are two possible readings of numeral-first $[-i]$ DPs: the cleft reading (with an indefinite pivot) and the definite reading. I have already given an analysis of how a cleft reading arises in a sentence with $[-i]$ in Bangla, when there is no numeral in it. Recall (32) and (27), repeated below from above:

- (32) a. ami e_i **ḡama-ṭa-i** por-b-o.
 1.SG.NOM this **dress-CLF-FOC** wear-FUT-1
 “It’s this dress that I’ll wear. (Not something else.)”
- b. $\llbracket \text{ami e}_i \text{ ḡama-ṭa-i por-b-o} \rrbracket$
 $= \begin{cases} 1 & \text{if the speaker wears the dress involved and only the dress involved} \\ 0 & \text{if the speaker doesn't wear the dress involved} \\ \# & \text{otherwise} \end{cases}$

$$\begin{aligned}
 (27) \quad & \llbracket -i^{st} \phi \rrbracket \\
 &= \begin{cases} 1 & \text{if } \llbracket \text{EXH}^{st} \phi \rrbracket = 1 \\ 0 & \text{if } \llbracket \neg \phi \rrbracket = 1 \\ \# & \text{if } \llbracket \phi \rrbracket = 1 \wedge \llbracket \text{EXH}^{st} \phi \rrbracket = 0 \end{cases} \\
 &\rightsquigarrow \begin{pmatrix} \text{prs: } \llbracket \phi \rrbracket = 1 \rightarrow \llbracket \text{EXH}^{st} \phi \rrbracket = 1 \\ \text{asr: } \llbracket \phi \rrbracket \end{pmatrix}
 \end{aligned}$$

That is, the analysis for the cleft reading is that the $[-i]$ morpheme attaches to the whole sentence and strengthens it. In other words, the Bangla sentence schematized as (33a) can be parsed as (33b), in which case the cleft reading arises. I’ve given the truth conditions for the LF in (33b) in (34). This is the cleft reading that $[-i]^{st}$ predicts, and there is nothing

surprising here, given the meaning I have assumed for clefts and $[-i]^{st}$. It mirrors exactly the meaning in (32b).

- (33) a. Mary read two books- $[i]$.
 b. $[-i]^{st} [[\exists [\text{two book PL}]] \lambda_1 \text{ Mary read } t_1]]$
 c. $[[\exists [-i^{et} [\text{two book PL}]]] \lambda_1 \text{ Mary read } t_1]$

$$(34) \llbracket (33b) \rrbracket = \begin{cases} 1 & \text{if Mary read two books and no more than two books} \\ 0 & \text{if Mary didn't read two books} \\ \# & \text{otherwise} \end{cases}$$

What I now propose — as will shortly seen to be in line with Doron (2025) — is that Bangla sentences like (33a) are actually structurally ambiguous between (33b) and (33c), where (33c) is the source of the definite reading. In other words, the definite reading also involves a cleft, but one that is embedded in the restrictor of the indefinite. $[-i]$ is type-flexible: it can attach to proposition-type nodes ((27), shown before), giving rise to the cleft reading, and it can attach to predicate-type nodes as well ((35)), giving rise to the definite reading, as I will explain below. The $\langle e, t \rangle$ -type-compatible variant of EXH is given in (36). In (35), C is the set of alternatives to P . (I am trying to keep the types as simple/extensional as possible; so I have put the evaluation index in the parameter.)

$$(35) \llbracket [-i^{et} P_{et}]^w = \lambda x_e . \begin{cases} 1 & \text{if } \llbracket \text{EXH}^{et} P_w \rrbracket(x) = 1 \\ 0 & \text{if } P_w(x) = 0 \\ \# & \text{otherwise} \end{cases}$$

$$(36) \llbracket \text{EXH}^{et} \rrbracket^w(C)(P_{et}) = \lambda X_e . \llbracket \text{EXH}^{st} \rrbracket^w(C)(P_w(X)) = 1$$

Since I have already demonstrated how the LF in (33b) composes, what remains to be seen is how (33c) helps derive the definite reading. The $[-i]$ in (33c) strengthens the truth condition of its prejacent, and, in the falsity condition, just negates the prejacent. The following is how it happens. The prejacent of $[-i]^{et}$ is “book PL”. Assuming the numeral “two” denotes the predicate of all those individuals which consist of exactly two atomic subparts, the meaning of this prejacent is the predicate of sums of books the cardinality of which is exactly 2. By *sum of books*, I mean any atomic or non-atomic entity consisting of book parts.

$$(37) \llbracket \text{two book PL} \rrbracket = \lambda X_e . X \text{ is a sum of books} \wedge |X| = 2$$

When this predicate is exhaustified by $[-i]^{et}$, we get the trivalent meaning in (38). To understand how this meaning comes about, one needs to ask, as in all cases of focus-sensitive operators like $[-i]$: what is the focus associate of $[-i]$; that is, what is the semantic pivot of

$[-i]$ (see section 3.2). My suggestion is to assume that the focus associate is the subject of the predicate. If this is assumed, then the result of strengthening in the truth condition ends up as the predicate of only those things which contain as their subparts only and all sums of books, that is, the singleton set of the maximal sum of books. The result of this is uniqueness/maximality. For instance, when there are three books, say, a , b , and c , then there is no sum of books with cardinality 2 that contain as their subparts only and all sums of books, because, here, the set of such things is $\{a \oplus b, b \oplus c, a \oplus c\}$. But when there are exactly two books, such as, a and b , this set becomes $\{a \oplus b\}$, whose only member *is* such a sum of books. This is exactly the partial function-like behavior we want to capture.

$$(38) \quad \llbracket [-i^{et} [\text{two book-PL}]] \rrbracket = \lambda X_e. \begin{cases} 1 & \text{if } X \text{ is a sum of books} \\ & \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2 \\ 0 & \text{if } X \text{ is not a sum of books} \vee |X| \neq 2 \\ \# & \text{otherwise} \end{cases}$$

$$\rightsquigarrow \lambda X_e. \begin{cases} \text{prs: } (X \text{ is not a sum of books} \vee |X| \neq 2) \\ \quad \vee (X \text{ is a sum of books} \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2) \\ \equiv \neg(X \text{ is not a sum of books} \vee |X| \neq 2) \\ \quad \rightarrow (X \text{ is a sum of books} \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2) \\ \equiv X \text{ is a sum of books and } |X| = 2 \\ \quad \rightarrow (X \text{ is a sum of books} \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2) \\ \equiv \text{if } X \text{ is a sum of books with two atoms,} \\ \quad \text{then } X \text{ is the maximal sum of books with two atoms} \\ \text{asr: } X \text{ is a sum of two books} \end{cases}$$

As I explained in section 3.1, in the trivalent framework, the felicity condition of any expression of any type is the disjunction of its truth and falsity conditions. Therefore, the predicate in (38) will yield a classical truth value only when the disjunction of its truth and falsity conditions are satisfied. This disjunction is fully spelled out above — first as a disjunction, then, through *modus tollens* and de Morgan’s law, as the conditional statement. This conditional statement can be described as the presupposition of this whole predicate.

When this trivalence projects to the matrix level according to the Strong Kleene schema in (39), we end up getting the presupposition in (40). The subscripted π is supposed to indicate that the predicate P comes with the presupposition π . That is, π is the conditional statement derived in the expanded “prs” in (38), P is the “asr” part, and Q is the nuclear scope. Although the result of this projection looks quite different from what is recognized as the uniqueness/maximality presupposition of a definite, it gives us an adequate result, in fact, a better result (see below), despite the DP being an indefinite. This captures the fact that a definite-like behavior is observed in numeral-first DPs in Bangla, which, without the cleft morpheme, can only be interpreted as an indefinite.

$$(39) \quad \exists x[P_\pi(x) \wedge Q(x)] = \begin{cases} 1 & \text{if } \exists x[\pi(x) \wedge P_\pi(x) \wedge Q(x)] \\ 0 & \text{if } \forall x[\neg Q(x) \vee [\pi(x) \wedge \neg P_\pi(x)]] \\ \# & \text{otherwise} \end{cases}$$

$$(40) \quad \llbracket [\exists [-i^{et} [\text{two book PL}]]] \lambda_1 \text{ Mary read } t_1 \rrbracket \\ = \begin{cases} 1 & \text{if } \exists X[X \text{ is a sum of books} \wedge |X| = 2 \wedge \text{Mary read } X] \\ & \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2] \\ 0 & \text{if } \forall X[X \text{ is not a sum of books} \vee |X| \neq 2 \vee \text{Mary didn't read } X] \\ \# & \text{otherwise} \end{cases}$$

These truth and falsity conditions have some welcome consequences, and also some unorthodox ones. Let us first see what it says about the standard uniqueness/maximality presupposition of definite descriptions. It basically says that the LF in (33c) is true if the predicate $\llbracket \text{two book-PL} \rrbracket$ has a unique maximal sum of books and Mary read it, it is false if there is no such entity that is also read by Mary. This is indeed the meaning we want for a sentence containing a definite DP. So, this is a good result. And in so being, it provides evidence for the claim about definites made in Doron (2025). I will now turn to the broader implications of the predicted presupposition.

6 Existence and proviso strengthening

The above theory of definites predicts something unorthodox when the presupposition of the whole LF in (33c) is considered. This becomes apparent when the felicity conditions in (40) are restated through the steps shown in (41). Recall that, in the trivalent framework, the presupposition/felicity condition of a sentence is the disjunction of the truth and falsity conditions of the whole sentence. This disjunction is shown in the first line in (41), for (40). After the restatements, the result is the conditional statement derived in the end. That is, the presupposition of the whole sentence is this conditional.

$$(41) \quad \begin{aligned} & \forall X[X \text{ is not a sum of books} \vee |X| \neq 2 \vee \text{Mary didn't read } X] \\ & \quad \vee (\exists X[X \text{ is a sum of books} \wedge |X| = 2 \wedge \text{Mary read } X] \\ & \quad \quad \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2]) \\ & \equiv \neg \forall X[X \text{ is not a sum of books} \vee |X| \neq 2 \vee \text{Mary didn't read } X] \\ & \quad \rightarrow (\exists X[X \text{ is a sum of books} \wedge |X| = 2 \wedge \text{Mary read } X] \\ & \quad \quad \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2]) \\ & \hspace{15em} (\textit{modus tollens}) \\ & \equiv \exists X \neg [X \text{ is not a sum of books} \vee |X| \neq 2 \vee \text{Mary didn't read } X] \\ & \quad \rightarrow (\exists X[X \text{ is a sum of books} \wedge |X| = 2 \wedge \text{Mary read } X] \\ & \quad \quad \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2]) \\ & \hspace{15em} (\neg \forall x \phi(x) \equiv \exists x \neg \phi(x)) \\ & \equiv \exists X[X \text{ is a sum of books} \wedge |X| = 2 \wedge \text{Mary read } X] \end{aligned}$$

$$\begin{aligned} &\rightarrow (\exists X[X \text{ is a sum of books} \wedge |X| = 2 \wedge \text{Mary read } X] \\ &\quad \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 2]) \\ &\hspace{15em} (\text{by de Morgan's law}) \end{aligned}$$

- ≡ If Mary read two books, then Mary read only two books
- ≡ If Mary read two books, then there are only two books

This is much weaker than what is usually said to be the presupposition that a definite DP is responsible for. That is, for instance, for the sentence in (42), a very weak presupposition is predicted, which is *If Mary read two books, then there are only two books*. This does not predict that the common ground would need to entail the existence of exactly two salient books. For instance, the sentence is predicted to be true when there are no books. This is not the standardly assumed presupposition of *the two books*, because (42) is usually taken to presuppose that there are exactly two books, and, in fact, if there aren't exactly two books, then the sentence is taken to be undefined. So, what does Doron's alternative route to definiteness have to say about this?

- (42) Mary didn't read the two books.

Doron (2025) has a discussion on this issue, where he attempts to justify the conditional presupposition. For the purposes of this paper, I would state my view on this in the following way. I believe that this issue — of a weaker presupposition being predicted than is usually perceived — is connected to the more general issue of proviso strengthening. It is often observed to be the case that when sentences with weaker presuppositions are uttered, stronger ones are accommodated, famously known as the proviso problem (Geurts 1996, 1999, Schlenker 2011, Singh 2006, 2007, 2008, 2009, among others). For instance, (43a) is predicted to have the conditional presupposition *If John flies to Toronto, then he has a sister*, but this is not what gets accommodated when this sentence is uttered; instead, the stronger proposition *John has a sister* is accommodated. However, the same thing is not observed in the case of (43b). In this case, the weaker, conditional presupposition *If John is a scuba diver, then he has a wetsuit* is much easier to accommodate, and the stronger proposition *John has a wetsuit* doesn't have to be accommodated.

- (43) a. If John flies to Toronto, then he's going there to meet his sister.
Semantic presupposition:
 If John flies to Toronto, then he has a sister.
Easily accommodated inference:
 John has a sister.
Inference much more difficult to accommodate:
 If John flies to Toronto, then he has a sister.
- b. If John is a scuba diver, then he'll bring his wetsuit.
Semantic presupposition:

If John is a scuba diver, then he has a wetsuit.

Easily accommodated inference:

If John is a scuba diver, then he has a wetsuit.

Inference that doesn't have to be accommodated:

John has a wetsuit.

(Singh 2008, based on (15), (16): 108-109)

How this is accounted for is not directly relevant for my purposes here. What is important for me is that what is hard to accommodate in (43a) can be made easy to accommodate, when the common ground makes the conditional presupposition plausible/likely. For instance, suppose we live in a world where Toronto is a terrible place to be for some reason, so there is no way anyone's ever traveling there unless forced to; and there was recently an earthquake in Toronto which resulted in mass fatality, and the interlocutors know that John has no living relatives — except, probably, a sister; therefore, if John goes to Toronto, then the only reason for that could be that he has some relative there who he wants to visit, and since he has no living relative other than, probably, a sister, it must be a sister. That is, in such a context, if (43a) is uttered, the much weaker and the otherwise much less plausible conditional proposition *If John goes to Toronto, then he has a sister* becomes easier to accommodate.

Now, notice that, for (42), similar pattern is observable: the theory I am advocating here predicts a weaker, conditional presupposition (*If Mary read two books, then there are only two books*), but the standardly reported *perceived* presupposition is exactly the consequent of this conditional proposition (*There are exactly two books*). So, if the pattern seen in (42) is indeed an instance of the proviso problem, then there should also be a mitigating effect of context there, as we did for (43a). I believe exactly this can be shown. For instance, consider (44). Since any human being has exactly two biological parents, the common ground entails that if two biological parents are found for anyone, then they have exactly two biological parents. Once the common ground is ensured to entail this, B saying that they did not find anyone for this person or robot on the database and that therefore this must be a robot no longer sounds incoherent. The standard understanding of the presupposition of definite descriptions predicts B's utterance to presuppose that there are exactly two biological parents for the subject in question; but that would be at odds with the completely natural conclusion B makes right after: that it must be a robot.

- (44) **Context:** There's something fully anthropomorphic sitting in front of two interlocutors, A and B. This thing might be an actual human being or it might be a robot. A can't tell whether it's actually human. But there's a way of running through a database, searching for a human being's biological parents. Every human being's information is on this database. The following conversation takes place.

A: Is this a human or a robot?

B: I'm not sure. I didn't find the two biological parents, so I guess it's a robot.

The account advanced here makes better predictions in this regard. It predicts exactly the conditional presupposition we need to explain the felicity of B's utterance: that if B finds two biological parents, then there are exactly two biological parents. This would mean that, as far as the reported stronger presupposition of sentences with definite descriptions, that might very well be the result of proviso-like strengthening. This is not to say we now understand how this strengthening comes about, because we don't really have a concrete account of proviso strengthening at this point. But the take-away is that the solution we might eventually find for proviso strengthening might help us with definite descriptions in the future as well.

An anonymous reviewer raises a very crucial point here, which is that this weaker conditional presupposition of definites that (44) seems to give us evidence for does not fully generalize. For instance, it has nothing to say about why the discourse in (45) is infelicitous. If the theory advocated here is right, then it should be felicitous, because it predicts that *Mary didn't read the two books* presupposes *If Mary read two books, there are exactly two books*, which is exactly what the first sentence asserts and is therefore expected to become common ground, thus leaving no apparent explanation for the unacceptability of the second sentence. This datapoint has indeed been one big reason for positing the canonical uniqueness/maximality presupposition for definites.

- (45) #If Mary read two books, there are exactly two books. Mary didn't read the two books.

In other words, Doron's theory predicts a weaker, conditional presupposition for both (44) and (45); but only in the former can this weaker presupposition be detected, and in the latter, the consequent of the condition — that is, something strictly stronger than the predicted presupposition — is perceived. The question, then, is why.

This is not an isolated problem, but a broader phenomenon that is observed when it comes to conditional presuppositions in general, and is thus, in general intertwined with the proviso problem. This point has been belabored in [Mandelkern \(2016\)](#), who makes the empirical observation that there are cases where even when the weaker, conditional presupposition predicted by the rules of presupposition projection is perfectly plausible given a certain contexts, speakers tend to favor the accommodation of the consequent of the conditional, that is, something strictly stronger than the predicted presupposition. This seems to happen apparently for no rhyme or reason. This is seen in (46), which is Mandelkern's. The paper has many other similar examples.

- (46) [It is common ground that Smith has gone missing, and we don't know whether he is still alive. A detective enters and says:] if the butler's clothes contain traces of Smith's blood, then it was the butler who killed Smith.

([Mandelkern 2016](#), (11): 8)

Most current theories of presupposition projection predict that sentences of the form (47a) trigger the presupposition of the form in (47b).

- (47) a. if p, then q.⁸
 b. *Semantic presupposition:*
 if p, then r.

If that is the case, though, then the predicted semantic presupposition of (46) is *If the butler's clothes contain traces of Smith's blood, then someone killed Smith*. However, this is not what the speaker of (46) is perceived to presuppose. Instead, just the consequent of this predicted presupposition, that is, something strictly stronger than it, is perceived as their presupposition. This has the exact same structure as the strengthening in (43a), where the predicted semantic presupposition was *If John flies to Toronto, then he has a sister*, while the perceived presupposition is *John has a sister*. But there is a crucial difference between the two cases. In (43a), the fact that conditional presupposition is hard to accommodate can be explained by saying that accommodating it results in a highly implausible common ground. But, one principal contribution of (46) is that it is specifically designed to make such a common ground highly plausible: that is, when the conditional presupposition is accommodated, the resulting common ground entails a link between traces of blood on the butler's clothes and Smith's murder, which is more than plausible. But still, it is the unconditional proposition that someone killed Smith that is perceived to be the presupposition. Mandelkern's point is: why should this be the case, if plausibility considerations are what leads to proviso strengthening in (43a)?

This is a worthy point, and I would conjecture that the question raised by (45) is subsumed by such cases of unexpected strengthening. In (45), it is not just plausible, but outright asserted, that there is a link between how many books Mary reads and how many books exist. But still, the following sentence containing *the two books* is infelicitous because it is perceived to presuppose, not something as weak as *If Mary read two books, then there are exactly two books*, but something strictly stronger than that, that is, *There are exactly two books*. This answer is admittedly unsatisfactory, but it is only so to the extent that the standard response to proviso strengthening involving plausibility is unsatisfactory.

Pertinently, the issue of the proviso problem seems to correspond exactly to the cases where local accommodation has been said to kick in (Heim 1990), for instance, (48). The standard story for this is that the existence presupposition gets locally accommodated under negation, which means that the inference from the first sentence is *It is not the case that there are exactly two books and Mary read them*. This is consistent with there being no books, which is why the whole discourse does not sound infelicitous. But notice that no such local accommodation remains necessary under the account advocated here. This is because the first sentence is predicted to presuppose *If Mary read two books, then there are exactly two books*, which does not entail the existence of exactly two books, thereby making it true when there are no books, which can thus be added in a second sentence without incurring

8. I am assuming Mandelkern's way of schematizing sentences with their presuppositions. That is, theoretically predicted semantic presuppositions of atomic presupposition bearers are subscripted to the propositions they are predicted to be presuppositions of. Italicized letters stand for propositions (metalanguage) and non-italicized letters stand for sentences (object language).

infelicity, as shown.

- (48) Mary didn't read the two books, because there are no books.

Whether this is the right prediction is a very subtle matter to navigate. But, here's one possible way of thinking about this (due to Amir Anvari, p.c.). Consider (49). If it was local accommodation that was responsible for the acceptability of *The King of France doesn't exist*, then its meaning, upon local accommodation, would be *It is not the case that there is a unique King of France*, which would license the disjunctive inference *Either France has no King or multiple kings*. But in that case, there will not be any explanation for why (49b) is not acceptable, while (49a) is. The putative disjunctive presupposition Doron's account predicts, *If France has a king, then there is exactly one King of France*, explains this, while local accommodation cannot. I would take this as an initial sign that Doron's approach is on the right track.

- (49) a. The King of France doesn't exist. That is, France has no king.
b. # The King of France doesn't exist. Actually, France has two kings.

7 Other advantages: “relative” and “absolute” definites

In this section, I want to present some entirely independent arguments for Doron's analysis of definites, which I am advocating here. The observations I will need to make are completely parallel to a much more familiar phenomenon that has long been observed in the context of superlatives. It is the distinction between absolute and relative superlatives. I will present this ambiguity in superlatives first, and then I will present the novel data I want to point out because the background on superlatives will help us orient ourselves in discussing definites in general. To preview what I am going to say, I will argue that all definites — not just superlative ones — are ambiguous between an absolute and a relative reading and this ambiguity receives a clean account if definites are always indefinites plus something else (e.g., CLEFT). These observations are totally independent of issues Doron is concerned with. My contribution here is naturally extending his analysis to capture these cases.

7.1 Background on relative and absolute superlatives

Superlatives have long been known to have an ambiguity between an absolute and a relative reading (see Heim 1999, Sharvit 2015, Sharvit & Stateva 2002, Szabolcsi 1986, among others). For instance, in Context 1 in (50a) makes the absolute reading of (50) salient and Context 2 in (50b) makes the relative reading of (50) salient. In the absolute reading, the comparison is “absolute”, that is, among all objects that meet the description of the superlative DP. For instance, in (50a), the comparison happens among all existing mountains; therefore, in the absence of additional context, in order for the sentence to be true, John must

have climbed Mount Everest. In the relative reading, comparison happens relative to a set that is made contextually salient. For instance, in (50b), the context makes salient the set of mountains climbed by John and a bunch of other people who are themselves contextually salient. The comparison takes place relative to this much smaller set than the set of all existing mountains. The relative reading in Context 2 can be made more accessible by putting focus on *John*. In a different context, something else could be focused too. For instance, *John CLIMBED the highest mountain* would convey that, among the set of mountains that John did various things to, climbing is what he did to the highest of them, and he did other things to the rest of them.

(50) John climbed the highest mountain.

- a. **Context 1, the absolute reading:** John climbed Mount Everest. No other contextually salient person has ever climbed any mountain.
- b. **Context 2, the relative reading:** John climbed the highest mountain in the set of mountains climbed by John and a bunch of other contextually salient people. None of them has climbed Mount Everest.

The most obvious way to account for the relative reading of superlatives might seem to be domain restriction, that is, to make the superlative's comparison class sensitive to context, as happens in usual cases of quantifier domain restriction, for instance, with *every* (von Stechow 1994). That is, just as *Every student passed* can be understood to mean *Every student in my class passed* or *Every student in this room passed*, *John climbed the highest mountain* can be understood to mean *John climbed the highest mountain among the set of mountains climbed by some contextually salient people*. Analyses of this kind are represented in Gutiérrez-Rexach (2006), Sharvit & Stateva (2000, 2002), Teodorescu (2009), Tomaszewicz (2015), a.o.

Another kind of account that has been proposed is one in which the superlative morpheme is interpreted outside the superlative DP, from where it has access to the focus structure of the entire section, allowing it to form the comparison class. Representatives of this approach include Hackl (2009), Heim (1999), Romero (2013), a.o. I will focus on Heim for the purposes of this paper. Her account is the following: if the superlative *-est* morpheme takes scope inside the superlative DP, then the absolute reading emerges, as shown in (51a), and if it scopes out of it, then the relative reading emerges, as shown in (51b). Her entry for *-est* is given in (52) for completeness, but it won't be directly relevant for me here. The point, schematically, is the following. For the absolute reading, comparison needs to happen among all existing mountains, which is ensured by the fact that, in (51a), the scope of "*C*₁ -est" is the predicate which denotes the relation that *x* bears to *d* if *x* is a *d*-high mountain. For the relative reading, the comparison needs to happen among only those mountains that have been climbed by a contextually salient set of individuals, that is, John and his competitors, which is ensured by the fact that, in (51b), the scope of "*C*₂ -est" is the relation that *x* bears to *d* if *x* climbs a *d*-high mountain. The effect of focus is captured by making *-est* focus-sensitive. So, when it has moved out of the superlative DP, it can associate with various

constituents in the structure. When it associates with *John*, the reading made salient by Context 2 emerges. When it associates with *climbed*, the other possible meaning I paraphrased right above (50) arises; *et cetera*.

(51) John climbed the highest mountain.

a. **LF for the absolute reading:**

John climbed [THE [C_1 -est] $\lambda d[d\text{-high mountain}]$]

$\rightsquigarrow$ *highest among all mountains*

C_1 = set of mountains

b. **LF for the relative reading:**

John [C_2 -est] $\lambda d[\text{climbed } [A\ d\text{-high mountain}]]$

$\rightsquigarrow$ *highest among the mountains climbed by one among several contextually salient individuals*

C_2 = set of contextually salient mountain-climbers

(Heim 1999, (30), (31): 13)

(52) $[-\text{est}] = \lambda C_{et} . \lambda R_{\langle d, et \rangle} . \lambda x . \exists d[R(d)(x) = 1 \wedge \forall y[y \neq x \wedge y \in C \rightarrow R(d)(y) = 0]]$

(Heim 1999, (10): 3)

One extremely crucial thing is revealed by the absolute-relative ambiguity in superlatives, which is the following. In the relative superlative reading of *John climbed the highest mountain*, it only makes sense to paraphrase the intended reading as *John, among a contextually salient set of individuals, climbed a mountain and the height of that mountain is greater than the mountains climbed by other members of this set*. Because there are multiple relevant mountains in the context, the object of *climbed*, semantically, must be an indefinite, despite the fact that *the* is pronounced. This has been a consistent problem for basically all accounts of the absolute-relative ambiguity. Usually, under these accounts, the necessity of an indefinite meaning in relative readings, despite the pronunciation of *the*, is stated as a stipulation (e.g., Heim 1999, Sharvit 2015, Szabolcsi 1986). There is currently no theory of superlatives, or definites in general, that can explain why the relative reading must be analyzed/paraphrased with an indefinite. This is important for me because, when I get to the problem I want to focus on below, an exactly parallel alternation between a definite and indefinite meaning is revealed, and that receives a natural explanation under my proposal about definites, because the crux of my proposal is that definites are always indefinites anyway, plus something else. Of course, the hope is that, at the end of the day, some unification is possible for my cases and the case of relative superlatives, through a general theory of definites; and that might, in fact, seem likely once I have pointed out the parallels. But I will not devote any space for that.

Before getting to the data crucial for me, one needs to be sensitive to the fact that relative readings of superlatives also arise when there is a bound pronoun inside the superlative, as

Bumford (2018) has pointed out. He presents data like (53). Here, the sentence contains the superlative expression *the oldest picture of his parents*. Like all superlatives, this also has two kinds of readings: absolute and relative. The absolute reading would be: of all pictures of John's parents, the oldest is in John's possession. This is not the reading that is relevant to the purpose at hand. The relevant reading is the relative reading of the superlative. In the relative reading, the object picked out by the superlative is neither the oldest existing picture of John's nor anyone else's parents; rather it's the oldest object in the set of objects in a smaller set. The ambiguity presented in (53) is about what that set is. When the reading in (53a) arises, it is the set of all the pictures of John's parents in the possession of John and a bunch of other people made salient by the context. When the reading in (53b) arises, it is the set of all those pictures such that it is a picture of their parents in their position (that is, children covarying with possessors).

(53) JOHN has the oldest picture of his parents.

- a. ***Relative reading, pronoun not bound by John:***
John has an older picture of John's parents than anyone else does of John's parents.
- b. ***Relative reading, pronoun bound by John:***
John has an older picture of John's parents than anyone else does of their parents.

(Bumford 2018, (9): 330)

Bumford presents (53) to argue that the domain restriction approach to the absolute-relative ambiguity does not work, because of specific problems introduced by binding. The details of his argument are not necessary for us to know here. All that is crucial for the following discussion is that superlatives are ambiguous between an absolute and relative readings; there is a domain restriction account and a scopal account of this ambiguity; and Bumford (2018) has argued, on the basis of binding data like (53) that the domain restriction account is inadequate.

This concludes the necessary background on superlatives. I will now present novel data to the effect that the absolute-relative ambiguity shown in (50) fully generalizes to all definites. Having cases like (50) in mind helps us see the exactly parallel ambiguity in definites in general. My crucial point will be that this more general ambiguity in definites can be given a clean account under my proposal about definites. Moreover, exactly as in the case of superlatives, a domain restriction account is inadequate for definites in general as well.

7.2 The facts: apparent “domain restriction” in definites

I will now present the data with definites which show an exactly parallel ambiguity to the one in (51). Let us begin with (54). The sentence in (54) is felicitous in Context 1. This

is expected from the conventional wisdom that definite descriptions come with a uniqueness/maximality presupposition, which is satisfied by the red ball John picks up in Context 1. However, on the face of it, this uniqueness/maximality presupposition is not satisfied in Context 2, because, as the context stipulates, there are multiple balls of each color. But still, the sentence is felicitous in this context.

(54) John picked the red ball.

a. **Context 1, absolute reading:**

There is a red ball, a blue ball, and a green ball on a table. John is about to pick one to play with and picks the red ball.

b. **Context 2, relative reading:**

There are some blue balls, red balls, and green balls in a jar. There are multiple balls of each color. As part of a game, John, Mary, and Bill put their hands in the jar and draw a ball without seeing which color they picked. So, it's up to chance what color they get and the idea is that depending on who gets which color, certain decisions in the game will be made. John gets red, Mary and Bill both get green.

Let us call the reading made salient by Context 1 *the absolute reading*, and the reading made salient by Context 2, *the relative reading*, because one could say that, in Context 2, the presupposition of the definite is satisfied *relative to* a certain set of balls, that is, the set of balls picked by John, Mary, and Bill. Again, just as in (50), focus has a central role to play here. A relative reading of (54) is made salient by Context 2, only if there is focus on *John*. But, in other contexts, different relative readings can emerge from different focus structure of the sentence. For instance, *John PICKED the red ball* would convey that, among the set of balls that John did various things to, picking is what he did to the unique red ball in that set, and he did other things to the rest of them. Notice the parallel with (50) in this respect, again.

The question remains, however, of how to formally analyze the reading of (54) made salient by Context 2. One obvious way to approach this question can be domain restriction, that is, intersecting the predicate *red ball* with a covert, pragmatically supplied predicate, with the meaning of “ $\lambda x_e . x$ is one of the balls picked out by John, Mary, and Bill”. The result of this intersection would be the singleton set containing the red ball that John picked out. Since this would be a singleton set, it would satisfy the presupposition of the definite determiner.

7.3 Problems with domain restriction

However, this kind of domain restriction analysis works only as long as there is no pronoun in the NP inside the definite, bound by something higher up, for instance, the subject. The completely parallel point to this is made by Bumford's (2018) (53) above, in the context of relative superlatives. With the discussion on superlative in the background now, in the exact same vein as in (53), consider (55). Observe, first of all, that the sentence is felicitous in the

given context. That means that *his* is not free; it's bound by *John*, and the intended meaning of the sentence is *John is the only individual x such that x took out a book by x's mother*. Second, let us say that there is a covert pragmatically supplied predicate meaning " $\lambda x_e . x$ is among the books that John, Mary, and Bill took out". When intersected with the predicate *book written by his mother*, the resulting predicate ends up being the set of the three books that John, Mary, and Bill took out. That is, firstly, this set is not a singleton set, leading to the violation of the uniqueness/maximality presupposition, and, secondly, there is no way for the co-varying reading to emerge.

(55) **Context:**

John, Mary, and Bill pick out books from the library to read. The mother of each is an author and their books are at the library they go to. John picks out a book written by his own mother, but Mary and Bill don't: they also pick out books written by John's mother. So, every book that is taken out is written by John's mother. There are other books written by their mothers in the library.

John took out the book written by his mother. (Mary and Bill didn't.)

Recall that, in the case of the absolute-relative ambiguity in superlatives, an alternation between the definite and indefinite meaning arises, when it comes to paraphrasing the meaning of the superlative (see just below (52)). The same definite-indefinite alternation arises in the case of (54) and (55). That is, it only makes sense to paraphrase the relative reading with an indefinite; it cannot be paraphrased using a definite. In (54), the relative reading must be paraphrased as *John picked a red ball and among the balls picked by John, Mary, and Bill, it is the only red ball*. In (55), the intended meaning is *John is the only individual x such that x took out a book x by x's mother*. This is despite the fact that both of the sentences make use of the definite determiner *the*. I will now propose that the ambiguity and the definite indefinite alternation seen in (50) should be given a unified analysis with those seen in (54). When this is done in the framework for definites I am advocating in this paper, none of this behavior is surprising and what is observed in the case of superlatives is a general property of definites that have gone unnoticed so far.

7.4 Moving the cleft operator

My account of the ambiguity in (54) is completely parallel to Heim's (1999) movement account of superlative semantics, with the added perk of capturing the alternation between definiteness and indefiniteness of the DP in question, depending on whether it is the absolute reading or the relative reading.

Recall Heim's scope account for the absolute-relative ambiguity of superlatives: if the superlative *-est* morpheme takes scope inside the superlative DP, then the absolute reading emerges, as shown in (51a), and if it scopes out of it, then the relative reading emerges, as shown in (51b). What I want to propose for (54) is virtually the same movement, but of the CLEFT operator I am proposing exists inside a definite DP, as in (56).

(56) John picked the red ball.

a. *LF for the absolute reading; ignoring QR of the indefinite:*

[John picked [$\exists$ [CLEFT [red ball]]]]

b. *LF for the relative reading; ignoring QR of the indefinite:*

[CLEFT [John picked [$\exists$ [CLEFT [red ball]]]]]



(56a) will be composed similarly to the cases of numeral-first [-i] definites of Bangla that I began with: the trivalence of “[CLEFT [red ball]]” will project out of the restrictor. For the same reasons uniqueness/maximality needs to be satisfied there — modulo the proviso issue — it will need to be satisfied here.

(56b) is the LF crucially responsible for the reading that is made salient by Context 2 in (54b). CLEFT moves in the narrow syntax from its DP-internal position to a propositional node. While, in (56a), CLEFT is LF-interpreted inside the DP — which is the source of the crucial aspect of the definiteness meaning — in (56b), it is LF-interpreted outside the DP that it was generated in. The effect of focus in (54) can be given a parallel account to the effect of focus in (50): CLEFT is a focus-sensitive operator and it can associate with various constituents in the structure.

There are two critical results of this movement that are relevant for us. First, the higher copy of CLEFT can now be interpreted in the higher position, from where it can associate with *John* and generate the set of alternatives. Second, now the quantifier is a plain indefinite, because CLEFT has moved out of it and, according to Doron’s theory, which I am advocating, CLEFT is the source of definiteness. Because the quantifier is now indefinite, the definite-indefinite alternation I discussed in section 7.3 is solved at once. Recall that the relative reading of (54) cannot be paraphrased using the definite determiner because there are multiple contextually salient red balls, and the only intuitive paraphrase of this reading is *John is the only individual who picked a red ball*, the object of *picked* being, crucially, an indefinite. There is nothing additional, let alone stipulative, that needs to be said about this under Doron’s theory: the DP has always been an indefinite anyway; it just looks like a definite because of DP-internal CLEFT semantics, and now that it has moved out of the DP, the indefiniteness becomes analytically obvious.

For the sake of completeness, the predicted meaning of the LF in (56b) will be the following. In the truth condition, because *John* is the semantic pivot of the cleft, John, as a red-ball-picker, is contrasted with everyone else. In the falsity condition, the prejacent of CLEFT is simply negated.

$$(57) \quad \llbracket (56b) \rrbracket = \begin{cases} 1 & \text{if John picked a red ball and no one else did} \\ 0 & \text{if John did not pick a red ball} \\ \# & \text{otherwise} \end{cases}$$

This semantic ambiguity between the absolute and the relative reading has no consequence for the PF, however. There, only the DP-internal copy of CLEFT is interpreted con-

sistently, regardless of which structure is sent off to PF by narrow syntax. Spell-out rules can then be written to the effect that “ $\exists$ CLEFT” strings are always pronounced as the definite determiner. This is why, under Doron’s theory, the sentence in (54) has two possible interpretations, but, because PF cannot see the distinctions that lead to them, there is no difference in pronunciation.

7.5 Szabolcsi (1986): Extraction with preposition stranding

The consequence of Doron’s proposal about definites is that definites lose their specific identity as definites, and become essentially indefinites. This idea may seem objectionable on the face of it, because there are environments where definites and indefinites behave differently. For instance, Szabolcsi (1986) has pointed out that extraction from an indefinite DP, stranding a preposition inside the DP, is allowed, but similar extraction from a definite DP is not allowed, as shown in (58). Given this asymmetry, Doron’s proposal can be charged with doing away with necessary differences between definites and indefinites, thereby making it harder to make sense of data like (58).

- (58) a. Who did you take a picture of?
b. *Who did you take the picture of?

(Szabolcsi 1986, (1): 1)

However, I would push against this charge and point out that whether or not (58b) is acceptable depends on the context in a way that is predicted by his proposal. For instance, it becomes acceptable in a context like the one in (59). The reason the sentence improves in this context is that this is a context that allows a relative reading of the definite to become salient, especially because *you* is focused. Here, the mother is understood to ask: *Who is the person x such that you are the only one who took a picture of x?* Crucially, because it is a relative reading, the paraphrase must involve an indefinite, and my extension of Doron’s proposal — involving movement of *cleft* — is perfectly able to capture this once again: through the movement of CLEFT out of the DP, then CLEFT associating with *you*, leaving a plain indefinite below. Because an indefinite is what is left below, the sentence becomes acceptable — exactly as one would predict given (58).

- (59) **Context:** Ann and Bill are siblings and they are both fans of a band. Their father takes them to a concert the band has and there, Ann and Bill manage to meet the members of the band. Ann takes a picture of one of the members and Bill takes a picture of another member. On their way back home, the father calls the mother to tell her everything that happened and how happy the kids are. So, the mother knows that the kids each took a picture of a different member of the band. When they come home, the mother asks Ann who she took a picture of. She answers. Then, she turns to Bill, and asks, with stress on *you*:

Who did YOU take the picture of?

Indeed, Szabolcsi's starting point was that, despite the asymmetry in (58), *Who did you take the best picture of?* is acceptable. The reason it is readily seen as acceptable is that the superlative, for some reason, inclines the hearer to favor a context where the relative reading is salient. In order to make such a context salient for a simple definite, as in (59), more effort needs to be made, for instance, through focus. The point, therefore, is *not* that (my extension of) Doron's proposal goes against Szabolcsi's (1986) insight about the distinction between definites and indefinites. The point, instead, is that the distinctions she pointed out between absolute and relative superlatives are so correct that they are not just about superlatives, but fully generalize to all definites. But explaining that requires an analysis that minimizes lexical differences between definites and indefinites, which has been achieved by Doron.

8 Coppock & Beaver's (2015) "Weak Uniqueness"

Essentially, what I am providing evidence for in this paper is a meaning of definite descriptions with conditional uniqueness, that is, uniqueness, provided something else is true. For instance, for the sentence *Mary has read the book*, the predicted presupposition is not *There is a unique book*, but something strictly weaker than it, that is, *If Mary has read a book, then there is exactly one book*. This predicted presupposition can be described as *weak uniqueness*. Notably, in a series of work, Coppock and Beaver (2012a, 2012b, 2014, 2015) have proposed exactly this, but only for predicative definites, not argumental ones, and they propose a different route to the weak uniqueness presupposition. In this section, I will point out some distinctions between their proposal and the one defended in this paper, and why I think the latter is conceptually superior. To contain the discussion, I will concentrate only on Coppock & Beaver (2015).

Coppock & Beaver's (2015) most crucial datapoint is that definite descriptions in predicative positions do not always lead to a uniqueness or existence inference. If it did, then (60b) would be infelicitous, since the predicate *only author of Waverley* is an empty set, and *author of Waverley* is a non-singleton set. That is, although the felicity of (60a) can be made sense of under the conventional view of uniqueness in definites, that of (60b) cannot.

- | | | | |
|------|----|---|--------------|
| (60) | a. | Scott is the only author of <i>Waverley</i> . | [1 author] |
| | b. | Scott is not the only author of <i>Waverley</i> . | [> 1 author] |

(Coppock & Beaver 2015, (2): 380)

Coppock & Beaver's (2015) solution is to propose a weak lexical semantics for *the*, thereby allowing for the possibility of non-existence in cases like (60b). This lexical entry is given in (61). The weakness involves the presupposition of *the* being its argument be either the empty set or a singleton set. The possibility of the empty set allows for the case of non-existence.

(60a) remains unproblematic for this entry for obvious reasons. But it makes a crucial difference for (60b): this sentence is no longer undefined because the non-existence of the only author of *Waverley* is subsumed under the “ $|P| \leq 1$ ” condition.

$$(61) \quad \llbracket \text{the} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x_e : |P| \leq 1 . P(x) = 1$$

(Coppock & Beaver 2015, (50): 395; notation modified)

This is Coppock & Beaver’s (2015) proposal for definites in predicative positions, as in (60). The other point that they emphasize is that, in argumental positions, definites always presuppose existence. For instance, (62) comes with the inference that there was exactly one invited talk.

$$(62) \quad \text{Anna didn't see the only invited talk.} \quad (\text{Coppock \& Beaver 2015, (80b): 407})$$

Their proposal for argumental definites involves two components: (i) type-shifting, and (ii) ensuring existence (which (61) does not do because of the “ ≤ 1 ” condition). Type-shifting is required anyway, because the output of *the* in (61) is of type $\langle e, t \rangle$, and, in an argument position, definites need to be of type e . Therefore, they adopt Partee’s (1986) ι -operator, given in (63). Existence is ensured by stipulating undefinedness if there is no unique element satisfying the description of the predicate argument of ι .

$$(63) \quad \llbracket \iota \rrbracket = \lambda P_{\langle e, t \rangle} . 1 \text{ if there is a unique } x \text{ such that } P(x) = 1; \text{ undefined otherwise}$$

(Coppock & Beaver 2015, (84), (86): 408; rephrased for my purposes)

I completely sympathize with the intuition Coppock & Beaver (2015) begin with, when they present (60) as their motivation for weak uniqueness. But I disagree with them when they so readily accept that argumental definites come with an existence presupposition. This is simply because of all the discussion in section 6 that was necessary to make some sense of the difference between (44) and (45). (44) was the case where the use of a definite in . . . *I did not find the two biological parents* . . . did not give rise to an existence presupposition, as opposed to (45), where such an inference arises in *Mary did not read the two books*. In short, (62) patterns with (45), in that, in both of these cases, existence is inferred. But, as I argued in that section, these cases of existence inference can be characterized as unexpected cases of proviso strengthening. Therefore, making existence a necessary part of argumental definites removes the very real murkiness introduced by the contrast between (44) and (45). Put differently, attributing the existence presupposition of (argumental) definites to proviso strengthening introduces into the issue a vagueness independently observed in proviso strengthening, thereby giving a unified view.

I should also point out that Coppock & Beaver’s (2015) central datapoint, that is, (60), receives a natural account under my analysis, as long as some assumptions are made about

the semantics of *only* and the pronunciation of the definite determiner, fully in the vain of the PF considerations at the end of section 7.4. This account is essentially an account of relative definiteness, as discussed in section 7. Notice that the sentences in (60) can be paraphrased as shown in (64). Crucially, these paraphrases are coherent only with an indefinite determiner, exactly as in the case of the relative reading of (54) in (54b).⁹

- (64) a. Scott is the only author of *Waverley*.
 $\rightsquigarrow$ Scott is an author of *Waverley* and no one else is.
 b. Scott is not the only author of *Waverley*.
 $\rightsquigarrow$ It is not the case that Scott is an author of *Waverley* and no one else is.

Assuming *only* is type-flexible and can take both propositions and predicates as its argument,¹⁰ the LFs for (60) will be the ones in (65), exactly as in (56b). And again, just as in the case of (56b), this will not make any difference for PF, where the output will be the "∃ only" string. If the indefinite determiner is exponed as *the* when followed by *only* (and CLEFT; see section 7.4), the pronunciation will always be what is seen in (60).¹¹

- (65) a. **LF for (64a); ignoring QR of the indefinite:**
 [only [Scott [is [∃ [only [author of *Waverley*]]]]]]
 $\uparrow$
 b. **LF for (64b); ignoring QR of the indefinite:**
 [not [only [Scott [is [∃ [only [author of *Waverley*]]]]]]
 $\uparrow$

In fact, such a movement analysis of *only* makes a better prediction than Coppock & Beaver's (2015) analysis. It is not really correct to say that (62) necessarily gives rise to a uniqueness presupposition. The reason this inference seems inevitable is because, as hearers, we tend to accommodate a context like Context 1 in (66a), which ensures the existence of a unique invited talk. But the sentence can also be felicitously uttered in Context 2 in (66b). Here, crucially, there are multiple invited talks, but what the sentence can be understood as is the following praphrase: *Anna is not the only individual who saw an invited talk; Bill saw invited talks too*. Notice the indefinite determiner in the paraphrase again.

- (66) Anna didn't see the only invited talk.
Good continuation for Context 2: . . . Bill also saw invited talks.

9. Virtually the same proposal is made in Sharvit (2015).

10. I will not elaborate on the details of the semantics of *only* here. I am referring to what was proposed as the crosscategorical semantics of *only* in Rooth (1985).

11. This would explain why the indefinite determiner is so rare with *only* and why expressions like *an only child* has an idiomatic flavor. But this intuition needs to be made more concrete, which I will not pursue here.

- a. **Context 1, absolute reading:** Anna is attending a conference, where there is only one invited talk. It is too early in the morning and she was too tired the night before; so she misses it.
- b. **Context 2, relative reading:** There is a conference going on. It has several invited talks scheduled. Anna, Bill, and Cathy are attending the conference. The invited talks are all scheduled at the beginning of each day of the conference, which is very early in the morning. Anna and Bill are enthusiastic early-risers, while Cathy likes sleeping in. So, Anna and Bill go to all the invited talks, and Cathy goes to none of them. Therefore, Anna is not the only individual who sees an invited talk, because Bill sees the invited talks as well.

Under my account, there is a way to account for the reading of (66) made salient by (66b): *only* is moved in narrow syntax and it then associates with *Anna*, leaving the indefinite determiner below. This is completely parallel to (56). But under Coppock & Beaver's (2015) account, (66) necessarily has a uniqueness presupposition because of how the *ι*-operator is interpreted (see in (63)); therefore, the sentence is wrongly predicted to be infelicitous in Context 2. To be fair to Coppock & Beaver (2015), they do mention that the uniqueness inference of (66) arises "in the absence of further supporting context" (*ibid.*, p. 414); but they do not end up pursuing the effect of context they seem to be hinting at here.

9 Conclusion and a wrinkle

The standard wisdom about definite descriptions is that they have a uniqueness/maximality presupposition. Recently, Omri Doron has been exploring a new way of deriving this presupposition, which involves presuppositionally exhaustifying the restrictor of an existential/indefinite quantifier and then letting that presupposition project from the restrictor to the root node following a Strong Kleene schema. This, then, shows the signature of what is usually recognized as the uniqueness/maximality presupposition. If this is indeed a possible route to this presupposition, then a natural prediction of this is that, in a language where we find an overt version of the presuppositionally exhaustifying operator, an indefinite DP should behave like a definite DP when that operator is added to the restrictor of the DP. I argue in this paper that Bangla is exactly such a language. In this language, the cleft morpheme — analyzable as the presuppositionally exhaustifying operator — when added to the restrictor of an indefinite DP with a numeral, makes it behave like a definite description with that numeral. In the part of the paper that followed, I adduced additional data in favor of such an analysis of definites, which are fully parallel to similar data found in the case of superlatives.

So, I have shown that there is empirical evidence for breaking up a definite determiner into an existential determiner and a presuppositional exhaustification operator. One part of the evidence in favor of this comes from Bangla numeral DPs. But, given all the data discussed there, the reader would be right to wonder what this system predicts when the

numeral is one and whether that prediction is borne out. Let us see the prediction first. I will then then characterize it appropriately, and see if the corresponding Bangla sentence meets those characterizations.

So, the new LF that now has to be considered is in (67).

$$(67) \quad [[\exists [-i^{et} [\text{one book SG}]]] \lambda_1 \text{ Mary read } t_1]$$

Just as before, we predict the following felicity conditions for this LF:

$$(68) \quad \begin{aligned} & [[[\exists [-i^{et} [\text{one book SG}]]] \lambda_1 \text{ Mary read } t_1]] \\ & = \begin{cases} 1 & \text{if } \exists X[X \text{ is a sum of books} \wedge |X| = 1 \wedge \text{Mary read } X \\ & \wedge \forall y[y \text{ is a sum of books} \rightarrow y \sqsubseteq x] \wedge |X| = 1] \\ 0 & \text{if } \forall X[X \text{ is not a sum of books} \vee |X| \neq 1 \vee \text{Mary didn't read } X] \\ \# & \text{otherwise} \end{cases} \end{aligned}$$

So, here is our prediction for sentences with the numeral one: when there is just one book, the definite reading of the numeral-first *[-i]* DP is available. This is not a good prediction because *[æk-ta/-ɔ̃on NP-i]* “one-CLF NP-FOC” doesn’t behave like definites, when the diagnostics I used before are performed on them. For instance, in (69), *[æk-ɔ̃on mofila-i]* “one-CLF woman-FOC” cannot refer back to the woman just introduced, that is, it does not have a definite reading, and, in (70), *[æk-ta nak-e-i]* “one-CLF nose-LOC-FOC” cannot behave like an indefinite.

- (69) o **æk-ta** **boi-i** k^hũɔ̃-e pe-et^h-e.
 3.SG.NOM **one-CLF** **book-FOC** find-GER get-PERF-PRS.3
 “He has found ONE book.”
 “He has found the book.”

- a. **Context 1 (the cleft reading):** Amol has found some books to add to the optional reading list for his class. He was looking for two more appropriate books to add. He told the speaker about his plans. But later, he was having a hard time finding two such books. So, he considers adding just one more book, which he manages to find, and tells the speaker about this change in plans, without saying which book he found. The speaker now does not know which books Amol has finally got, but only that Amol has found a book that he can now add to his list of optional readings. So, he corrects his beliefs.
- b. # **Context 2 (the definite reading):** Amol and Charu are siblings and go the bookstore together to buy books. Amol gets one book. Charu is present during the purchase. She sees the book, and when they come home, their mother sees the book as well. After that, Amol is seen reading it for a few days. Then Amol loses the book. Amol, Charu, and their mother look for it together. After a long search, Amol manages to find it and Charu goes to their mother to report that, so she can stop looking.

- (70) a. #ama-r æk-ta nak-e g^hũfi k^he-et^h-i.
 1.SG-GEN **one-CLF** **nose-LOC** punch eat-PERF-PRS.1
 #“I got punched in a nose that’s mine.”
- b. #ama-r æk-ta nak-e-i g^hũfi k^he-et^h-i.
 1.SG-GEN **one-CLF** **nose-LOC-FOC** punch eat-PERF-PRS.1
Intended but unavailable reading that would have been felicitous:
 “I got punched in my nose.”

Although, in this respect, I make a problematic prediction, there are reasons to believe that, while I can successfully account for the plural cases, what happens in singular DPs with a numeral is an uncharted territory to begin with. This is because, while switching the order between the numeral and NP changes the DPs (in)definiteness property (see (9)), the same is not true for [æk-ta/-ɕon NP] “one-CLF NP”. In fact, in this case, only the numeral-first order is acceptable, and the NP-first order is simply impossible, as shown in (71). If the definite singular meaning is intended, one must drop the numeral altogether and say (71c).

- (71) a. ami æk-ta boi poṭ-et^h-i.
 1.SG.NOM **one-CLF** **book** read-PERF-PRS.1
 “I’ve read a book.”
- b. *ami boi æk-ta poṭ-et^h-i.
 1.SG.NOM **book** **one-CLF** read-PERF-PRS.1
Intended:
 “I’ve read the book.”
- c. ami boi-ta poṭ-et^h-i.
 1.SG.NOM **book-CLF** read-PERF-PRS.1
 “I’ve read the book.”

This means that neither the DP-internal NP movement past the numeral nor the definite reading of numeral-first [-i] DPs discussed in section 4 is observed when the numeral is one. Now, if this is because of the numeral being as high as a quantifier, it is still not explained why [æk-ta nak-e-i] “one-CLF nose-LOC-FOC” cannot have a definite reading, because it can still be parsed as (72), where “one-CLF” sits in a position to the left of where it sits in (67).

- (72) [one-CLF [-i^{et} [nose SG]]]

If it is parsed this way, then, the usual uniqueness/maximality presupposition of a definite should still arise. One possibility is to stipulate that the movement of [æk-ta] “one-CLF NP-FOC” to the edge of the DP satisfies whatever feature [-i] would have satisfied if it had been attached, which thus makes it impossible to be attached for featural reasons, thereby making the parse in (72) ungenerable. This, however, would not rule out the cleft LFs, where [-i] attaches at the matrix level, for instance, as in (73), for (70a).

(73) [-i [ama-r æk-ta nak-ei g^hũfi k^he-et^h-i]]

This would be a good result, because, although it is desirable to predict the definite reading to be unavailable in this specific numeral-first DP (because of (71b)), it is also desirable to predict its cleft reading to still be available (because of (69)).

Glossing abbreviations

1	1st person
2	2nd person
3	3rd person
CLF	Classifier
FOC	Focus
FUT	Future
GEN	Genitive
GER	Gerund
HAB	Habitual
IMPFV	Imperfective
LOC	Locative
NEG	Negation
NMLZ	Nominalizer
NOM	Nominative
PERF	Perfect
PFV	Perfective
PRS	Present
Q _{Y/N}	Question particle in Yes/No questions

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